



KAMBALA

Student Number: _____

**Yearly Examination
September 2019**

Preliminary Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A Reference Sheet is provided for use in this task
- Answer questions 1 – 10 on the answer sheet provided
- Answer questions 11 through 26 on the paper provided
- For questions 11 through 26, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt Questions 11 through 26
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt questions 1-10.

Allow about 15 minutes for this section.

Use the answer sheet for Questions 1 – 10.

1. If $\sqrt{a} + \sqrt{a} = 16$, what is the value of a ?

(A) $2\sqrt{2}$

(B) 8

(C) 16

(D) 64

2. What is the simplest form of $\frac{x^6}{y^6} \div x^2y^{-3}$?

(A) $\frac{x^3}{y^2}$

(B) $\frac{x^4}{y^2}$

(C) $\frac{x^4}{y^3}$

(D) $\frac{x^4}{y^9}$

3. What are the coordinates of the vertex of the parabola with equation $y = x^2 + 4x + 5$?

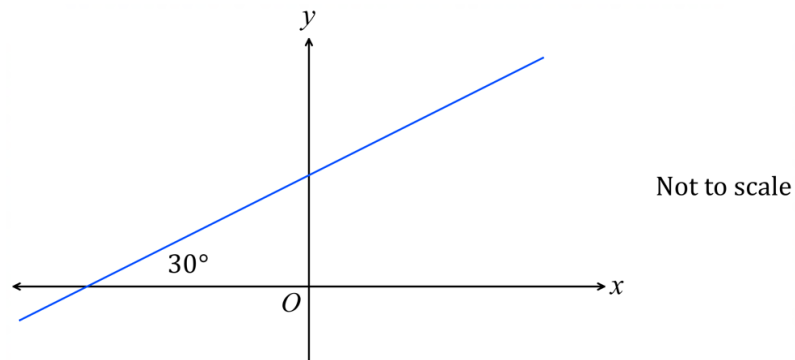
(A) $(2, -1)$

(B) $(2, 1)$

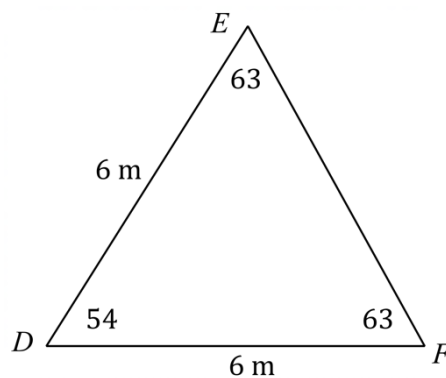
(C) $(-2, -1)$

(D) $(-2, 1)$

4. A line makes an angle of 30° with the positive direction of the x -axis. What is the gradient of the line?



- (A) $\frac{1}{\sqrt{3}}$
(B) $\frac{1}{\sqrt{2}}$
(C) $\frac{1}{2}$
(D) $\frac{\sqrt{3}}{2}$
5. What is the area of $\triangle DEF$?



- (A) 11.71 m^2
(B) 13.33 m^2
(C) 14.56 m^2
(D) 16.04 m^2

6. Which of the following functions is not one-to-one?

(A) $f(x) = 4 - x^2, x \geq 0$

(B) $f(x) = \sqrt{4 - x^2}$

(C) $f(x) = 1 - 4x$

(D) $f(x) = \frac{4}{x}$

7. What is the equation of the asymptote of the graph with equation $y = 4 \log_2(5x) + 2$?

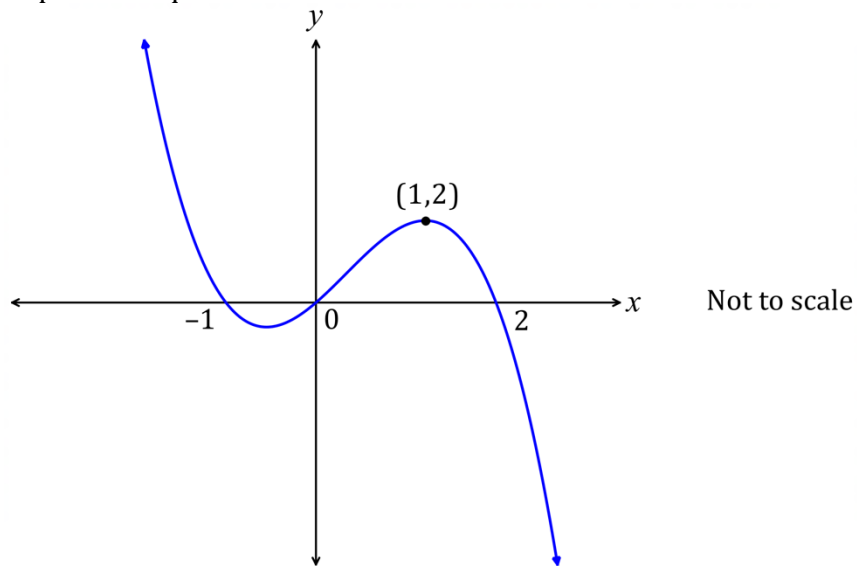
(A) $x = 0$

(B) $x = 2$

(C) $x = 4$

(D) $x = 5$

8. What is a possible equation of the cubic function below?



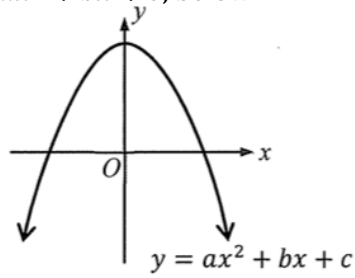
(A) $y = -x(x + 1)(x - 2)$

(B) $y = -x(x - 1)(x + 2)$

(C) $y = x(x + 1)(x - 2)$

(D) $y = x(x - 1)(x + 2)$

9. Consider the parabola $y = ax^2 + bx + c$, below.



Which of the following is true?

- (A) $b^2 - 4ac > 0$ and $a > 0$
 - (B) $b^2 - 4ac < 0$ and $a > 0$
 - (C) $b^2 - 4ac > 0$ and $a < 0$
 - (D) $b^2 - 4ac < 0$ and $a < 0$
10. What are the solutions of the equation $|x - 1| - 2 = 0$?
- (A) $x = -3$ or $x = 1$
 - (B) $x = 3$ or $x = -1$
 - (C) $x = -3$ or $x = -1$
 - (D) $x = 3$ or $x = 1$

Section II

60 marks

Attempt all questions.

Allow about 1 hour and 45 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of the examination paper.

- 11.** Rationalise the denominator of $\frac{2}{\sqrt{3}+1}$. **2**

- 12.** Find the exact value of $\cos \theta$ if $\sin \theta = \frac{2}{5}$ and $90^\circ < \theta < 180^\circ$. **2**

- 13.** Differentiate the following expressions with respect to x .

- (a)** $(3x^2 - 5x)^5$ **2**

(b) $2x^2 + 5 - \frac{7}{x}$ 2

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$$(c) \quad \frac{4x^3 + 5}{2x - 9} \qquad \qquad \qquad \mathbf{3}$$

[illegible]

14. Simplify:

(a) $\frac{1}{2} \log_x 16 + 2 \log_x 5$ **2**

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50</
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(b) $\frac{x + 2y}{3} - \frac{2x - y}{4}$ 2

(c) $\frac{(2x^4)^3}{5x^{-1}}$ 2

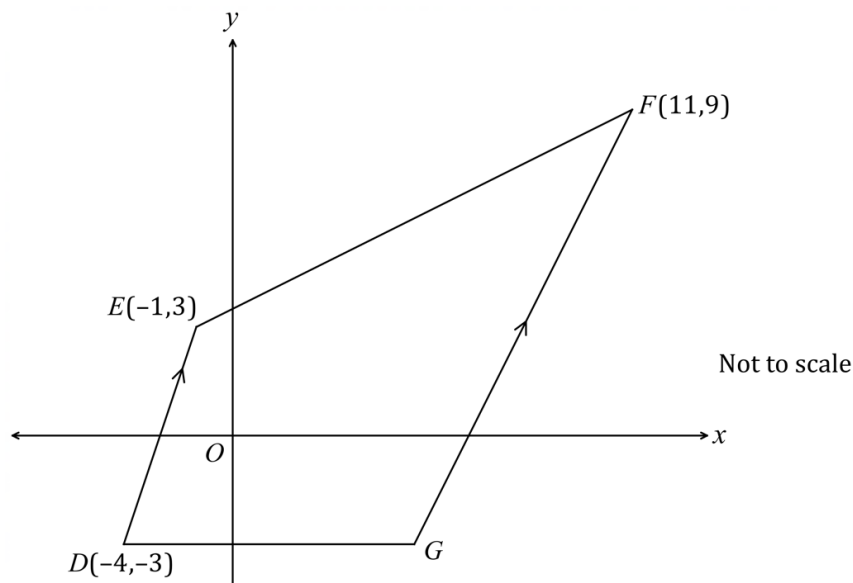
15. Given that $f(x) = x^2 - x$ and $g(x) = 2x + 1$, find:

(a) $f(-2)$ 1

(b) $f(g(x))$ 2

(c) x where $g(x) = 0$. 1

16.



On a number plane the points D , E and F have coordinates $(-4, -3)$, $(-1, 3)$ and $(11, 9)$ respectively. DE is parallel to GF and DG is parallel to the x -axis.

(a) What is the gradient of DE ?

1

(b) What is the midpoint of DE ?

1

(c) Find the equation of the line FG .

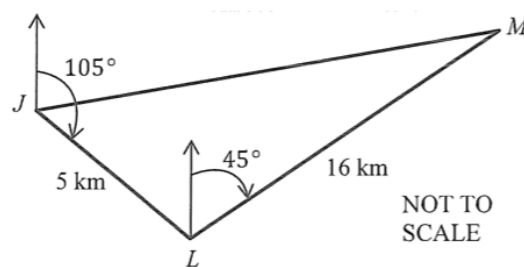
2

- 18.** Solve the equation $2 \sin x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

2

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19. Mia swims from the point J on a bearing of 105° for 5 km to the point L . She then changes direction and swims on a bearing of 045° for a further 16 km to the point M .



- (a) Find $\angle JLM$.

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- (b) Find the distance JM that Mia will swim back to the point J . **2**

[illegible]

20. Solve the equation $\log_2 \frac{x+1}{x-1} = 2$ 2

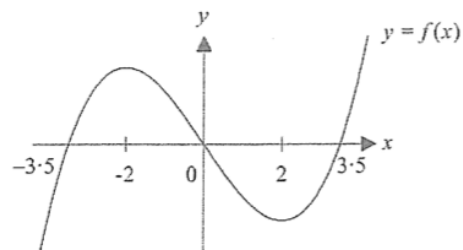
[illegible]

- 21.** Use the formula $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ when $f(x) = x^2 + 3$. **2**

[illegible]

- 22.** Show that $\frac{1+\cos x}{\sin x} + \frac{\sin x}{1+\cos x} = 2\operatorname{cosec} x$.

3



2

Sketch the curve $y = f'(x)$ showing the shape of the curve and the x intercepts.

24. Sketch the curve $y = \sqrt{x - 2} + 3$ and state its domain and range. 4

25. A water tank is being filled at a variable rate. The height of the water, h cm, at any time, t minutes can be described by $h(t) = 2t^2 - 4t + 50$. Find:

- (a) The initial height of the water in the tank. 1

- (b) The instantaneous rate at which the height is changing at 4 minutes. 2

- (c) The average rate at which the height has changed over the first 4 minutes. **2**

[illegible]

- 26.** The number of bats, N , in the park at time t years, is given by the formula $N = 40 \times 3^{0.327t}$.

- (a) How many bats were in the park initially? 1

[illegible]

- (b) Make t the subject of the formula. 2

[illegible]

- (c) How many years would it take for the number of bats to reach 80? Answer to the nearest year. 2

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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Student Number: _____

Preliminary Mathematics Advanced

**Yearly Examination
September 2019**

Section I

Multiple-Choice Answer Sheet
Circle the answer of your choice.

- | | | | | |
|-----|---|---|---|---|
| 1. | A | B | C | D |
| 2. | A | B | C | D |
| 3. | A | B | C | D |
| 4. | A | B | C | D |
| 5. | A | B | C | D |
| 6. | A | B | C | D |
| 7. | A | B | C | D |
| 8. | A | B | C | D |
| 9. | A | B | C | D |
| 10. | A | B | C | D |



KAMBALA

Student Number: Solutions

**Yearly Examination
September 2019**

Preliminary Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- A Reference Sheet is provided for use in this task
- Answer questions 1 – 10 on the answer sheet provided
- Answer questions 11 through 26 on the paper provided
- For questions 11 through 26, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt Questions 11 through 26
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt questions 1-10.

Allow about 15 minutes for this section.

Use the answer sheet for Questions 1 – 10.

1. If $\sqrt{a} + \sqrt{a} = 16$, what is the value of a ?

(A) $2\sqrt{2}$

(B) 8

(C) 16

(D) 64

$$2\sqrt{a} = 16$$

$$\sqrt{4a} = 16$$

$$4a = 256$$

$$a = 64$$

2. What is the simplest form of $\frac{x^6}{y^6} \div x^2y^{-3}$?

(A) $\frac{x^3}{y^2}$

(B) $\frac{x^4}{y^2}$

(C) $\frac{x^4}{y^3}$

(D) $\frac{x^4}{y^9}$

$$\frac{x^6}{y^6} \times \frac{y^3}{x^2} = \frac{x^4}{y^3}$$

3. What are the coordinates of the vertex of the parabola with equation $y = x^2 + 4x + 5$?

(A) $(2, -1)$

(B) $(2, 1)$

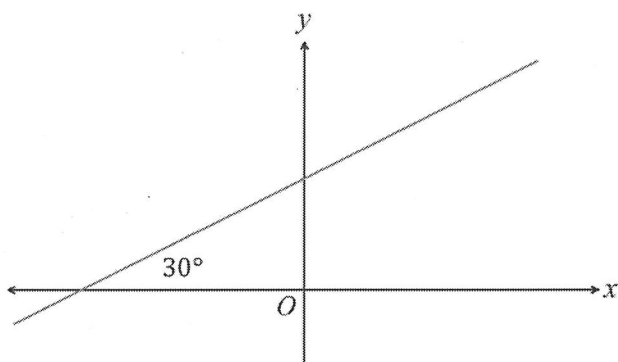
(C) $(-2, -1)$

(D) $(-2, 1)$

$$x = -\frac{4}{2} = -2$$

$$y = 4 - 8 + 5 = 1$$

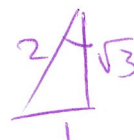
4. A line makes an angle of 30° with the positive direction of the x -axis. What is the gradient of the line?



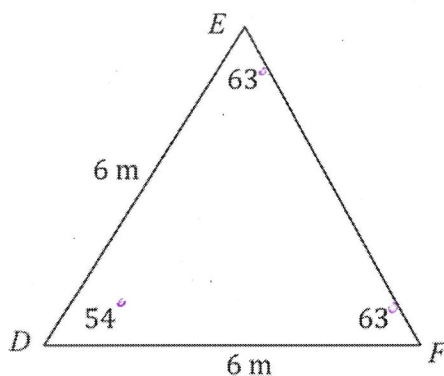
Not to scale

- (A) $\frac{1}{\sqrt{3}}$
 (B) $\frac{1}{\sqrt{2}}$
 (C) $\frac{1}{2}$
 (D) $\frac{\sqrt{3}}{2}$

$$\begin{aligned}\tan \theta &= m \\ m &= \tan 30^\circ \\ &= \frac{1}{\sqrt{3}}\end{aligned}$$



5. What is the area of $\triangle DEF$?



- (A) 11.71 m^2
 (B) 13.33 m^2
 (C) 14.56 m^2
 (D) 16.04 m^2

$$\begin{aligned}A &= \frac{1}{2} \times 6 \times 6 \times \sin 54^\circ \\ &= 14.56\end{aligned}$$

6. Which of the following functions is not one-to-one?

(A) $f(x) = 4 - x^2, x \geq 0$

(B) $f(x) = \sqrt{4 - x^2}$

(C) $f(x) = 1 - 4x$

(D) $f(x) = \frac{4}{x}$

7. What is the equation of the asymptote of the graph with equation $y = 4 \log_2(5x) + 2$?

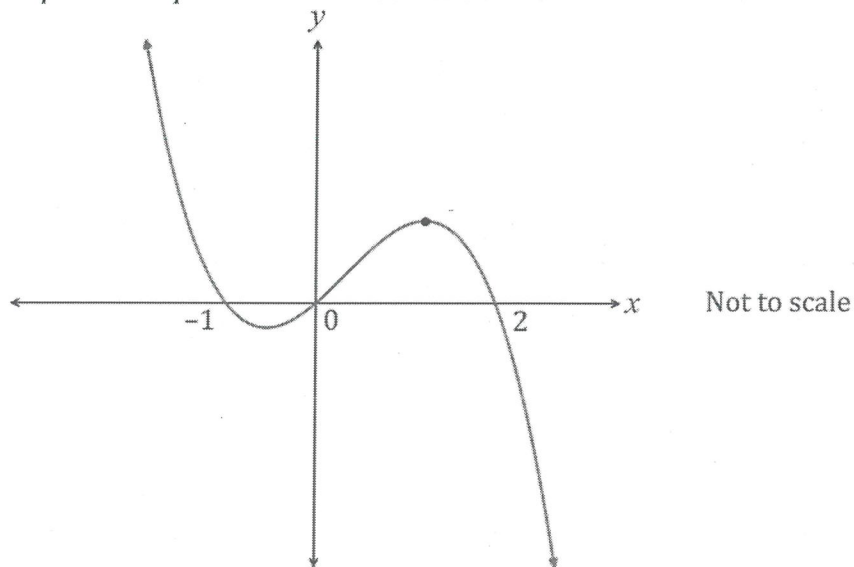
(A) $x = 0$

(B) $x = 2$

(C) $x = 4$

(D) $x = 5$

8. What is a possible equation of the cubic function below?



(A) $y = -x(x + 1)(x - 2)$

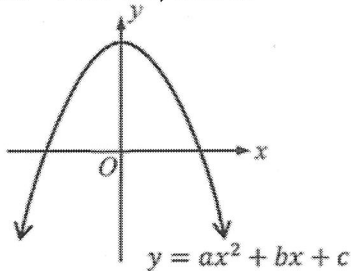
(B) $y = -x(x - 1)(x + 2)$

(C) $y = x(x + 1)(x - 2)$

(D) $y = x(x - 1)(x + 2)$

$-x(x+1)(x-2)$

9. Consider the parabola $y = ax^2 + bx + c$, below.



Which of the following is true?

- (A) $b^2 - 4ac > 0$ and $a > 0$
(B) $b^2 - 4ac < 0$ and $a > 0$
(C) $b^2 - 4ac > 0$ and $a < 0$
(D) $b^2 - 4ac < 0$ and $a < 0$

10. What are the solutions of the equation $|x - 1| - 2 = 0$?

- (A) $x = -3$ or $x = 1$
(B) $x = 3$ or $x = -1$
(C) $x = -3$ or $x = -1$
(D) $x = 3$ or $x = 1$

$$|x-1|=2$$

$$x-1=2$$

$$x=3$$

$$x-1=-2$$

$$x=-1$$

Section II

60 marks

Attempt all questions.

Allow about 1 hour and 45 minutes for this section.

Answer the questions in the spaces provided.

Your responses should include relevant mathematical reasoning and/or calculations.

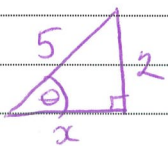
Extra writing space is provided at the back of the examination paper.

11. Rationalise the denominator of $\frac{2}{\sqrt{3}+1}$. 2

$$\frac{2}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{2\sqrt{3}-2}{2}$$

$$= \sqrt{3}-1$$

12. Find the exact value of $\cos \theta$ if $\sin \theta = \frac{2}{5}$ and $90^\circ < \theta < 180^\circ$. 2



$$\cos \theta = -\frac{\sqrt{21}}{5} \quad \text{as } 90^\circ < \theta < 180^\circ$$

$$x = \sqrt{25-4}$$

$$= \sqrt{21}$$

13. Differentiate the following expressions with respect to x .

- (a) $(3x^2 - 5x)^5$ 2

$$y' = 5(3x^2 - 5x)^4 (6x - 5)$$

(b) $2x^2 + 5 - \frac{7}{x} = 2x^2 + 5 - 7x^{-1}$ 2

$$y' = 4x + 7x^{-2} \quad \text{or} \quad y' = 4x + \frac{7}{x^2}$$

(c) $\frac{4x^3 + 5}{2x - 9}$ 3

$$u = 4x^3 + 5 \quad v = 2x - 9$$

$$u' = 12x^2 \quad v' = 2$$

$$y' = \frac{12x^2(2x-9) - 2(4x^3+5)}{(2x-9)^2} = \frac{24x^3 - 108x^2 - 8x^3 - 10}{(2x-9)^2}$$

$$= \frac{16x^3 - 108x^2 - 10}{(2x-9)^2}$$

14. Simplify:

(a) $\frac{1}{2} \log_x 16 + 2 \log_x 5$ 2

$$\log_x \sqrt{16} + \log_x 5^2 = \log_x 4 \times 25 = \log_x 100$$

(b) $\frac{x+2y}{3} - \frac{2x-y}{4}$ 2

$$= \frac{4(x+2y) - 3(2x-y)}{12}$$

$$= \frac{4x+8y-6x+3y}{12}$$

$$= \frac{-2x+11y}{12}$$

(c) $\frac{(2x^4)^3}{5x^{-1}}$ 2

$$\frac{8x^{12}}{5x^{-1}} = \frac{8}{5}x^{13}$$

15. Given that $f(x) = x^2 - x$ and $g(x) = 2x + 1$, find:

(a) $f(-2) = (-2)^2 - (-2)$ 1

$$= 4 + 2$$

$$= 6$$

(b) $f(g(x)) = f(2x+1)$ 2

$$= (2x+1)^2 - (2x+1)$$

$$= 4x^2 + 4x + 1 - 2x - 1$$

$$= 4x^2 + 2x$$

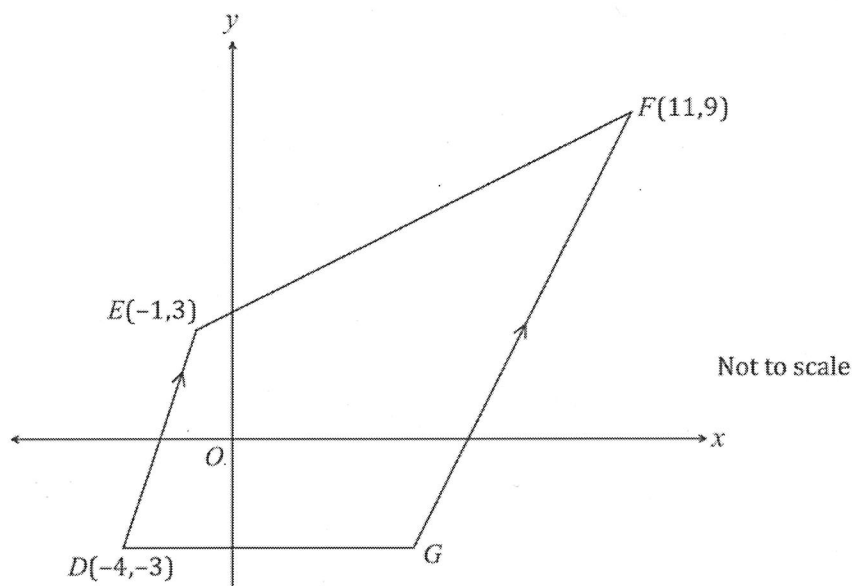
(c) x where $g(x) = 0$. 1

$$2x+1=0$$

$$2x=-1$$

$$x=-\frac{1}{2}$$

16.



On a number plane the points D , E and F have coordinates $(-4, -3)$, $(-1, 3)$ and $(11, 9)$ respectively. DE is parallel to GF and DG is parallel to the x -axis.

(a) What is the gradient of DE ?

1

$$m = \frac{3 - (-3)}{-1 - (-4)} = \frac{6}{3} = 2$$

(b) What is the midpoint of DE ?

1

$$x = \frac{-1 - 4}{2} = -\frac{5}{2} \quad y = \frac{3 - 3}{2} = 0$$

$$\left(-\frac{5}{2}, 0\right)$$

(c) Find the equation of the line FG .

2

$$y - y_1 = m(x - x_1)$$

$$y - 9 = 2(x - 11)$$

$$y - 9 = 2x - 22$$

$$y = 2x - 13$$

- (d) Show that the coordinates of the point
- G
- are
- $(5, -3)$
- .

2

 G is on the line FG

$$\therefore -3 = 2 \times 5 - 13$$

$$-3 = 10 - 13 \text{ correct } \therefore G(5, -3)$$

- (e) Find the distance
- EF
- .

1

$$EF = \sqrt{(11+1)^2 + (9-3)^2}$$

$$= \sqrt{121+36}$$

$$= \sqrt{157} \text{ or } 12.5$$

- (f) Find the equation of a circle with the centre at
- F
- and the radius
- EF
- .

1

$$(x-11)^2 + (y-9)^2 = 157$$

17. Find the equation of the tangent to the curve
- $y = \frac{1}{3}x^3 - 4x^2 + 8x + 3$
- at the point
- $P(3, m)$
- .

3

$$x=3 \quad y = \frac{1}{3} \times 27 - 4 \times 9 + 8 \times 3 - 3 = 0$$

$$P(3, 0)$$

$$y' = \frac{1}{3} \times 3x^2 - 8x + 8$$

$$\text{when } x=3 \quad m_t = 9 - 8 \times 3 + 8 = -7$$

$$y = -7(x-3)$$

$$y = -7x + 21$$

18. Solve the equation $2 \sin x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

2

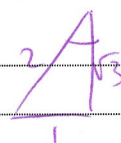
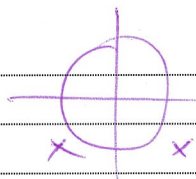
$$2 \sin x = -1$$

$$\sin x = -\frac{1}{2}$$

related angle = 30°

$$x = 180^\circ + 30^\circ, 360^\circ - 30^\circ$$

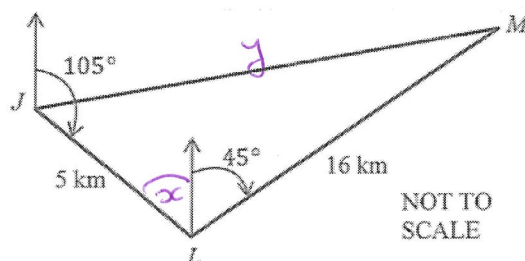
$$x = 210^\circ, 330^\circ$$



$$\sin x = \frac{1}{2}$$

$$x = 30^\circ$$

19. Mia swims from the point J on a bearing of 105° for 5 km to the point L . She then changes direction and swims on a bearing of 045° for a further 16 km to the point M .



- (a) Find $\angle JLM$.

1

$$x = 180^\circ - 105^\circ \text{ (co-interior angles on parallel lines)}$$

$$x = 75^\circ$$

$$\angle JLM = 75^\circ + 45^\circ$$

$$= 120^\circ$$

- (b) Find the distance
- JM
- that Mia will swim back to the point
- J
- .

2

$$y^2 = 5^2 + 16^2 - 2 \times 5 \times 16 \cos 120^\circ$$

$$y = \sqrt{25 + 256 - 160 \cos 120^\circ}$$

$$y = 19 \text{ km}$$

20. Solve the equation
- $\log_2 \frac{x+1}{x-1} = 2$

2

$$\frac{x+1}{x-1} = 4$$

$$x+1 = 4(x-1)$$

$$x+1 = 4x-4$$

$$3x = 5$$

$$x = \frac{5}{3}$$

21. Use the formula
- $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- to find
- $f'(x)$
- when
- $f(x) = x^2 + 3$
- .

2

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3 - x^2 - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3 - x^2 - 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h}$$

$$= 2x$$

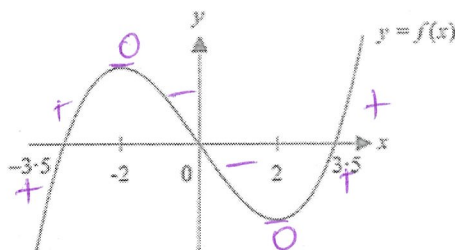
22. Show that $\frac{1+\cos x}{\sin x} + \frac{\sin x}{1+\cos x} = 2\operatorname{cosec} x$.

3

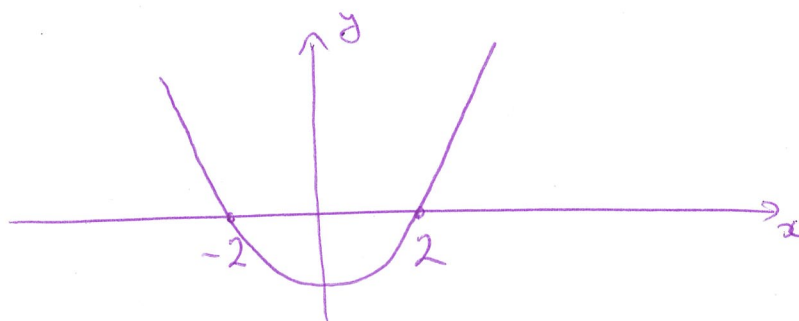
$$\begin{aligned}
 \text{LHS} &= \frac{(1+\cos x)^2 + \sin^2 x}{\sin x(1+\cos x)} \\
 &= \frac{1+2\cos x+\cos^2 x+\sin^2 x}{\sin x(1+\cos x)} \\
 &= \frac{2+2\cos x}{\sin x(1+\cos x)} \\
 &= \frac{2(1+\cos x)}{\sin x(1+\cos x)} \\
 &= \frac{2}{\sin x} \\
 &= 2\operatorname{cosec} x \\
 &= \text{RHS}
 \end{aligned}$$

- 23.

2

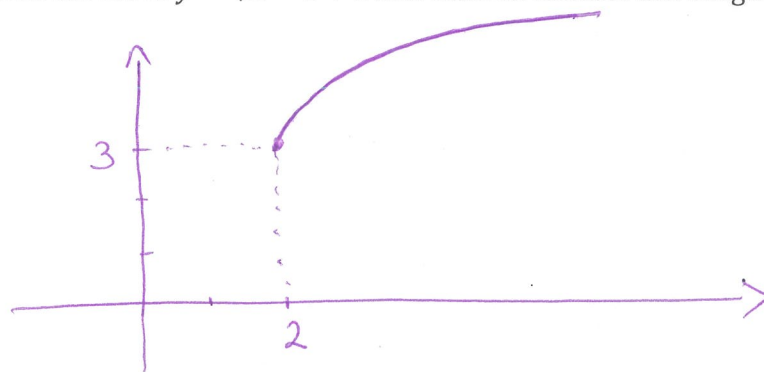


Sketch the curve $y = f'(x)$ showing the shape of the curve and the x intercepts.



24. Sketch the curve $y = \sqrt{x-2} + 3$ and state its domain and range.

4



$$x-2 \geq 0$$

$$x \geq 2$$

Domain: $x \geq 2$
Range: $y \geq 3$

25. A water tank is being filled at a variable rate. The height of the water, h cm, at any time, t minutes can be described by $h(t) = 2t^2 - 4t + 50$. Find:

- (a) The initial height of the water in the tank.

1

$$t=0 \quad h=50\text{cm}$$

- (b) The instantaneous rate at which the height is changing at 4 minutes.

2

$$\frac{dh}{dt} = 4t - 4$$

$$t=4 \quad \frac{dh}{dt} = 16 - 4 = 12\text{ cm/min}$$

- (c) The average rate at which the height has changed over the first 4 minutes.

2

$$t=4 \quad h = 2 \times 16 - 4 \times 4 + 50 \\ = 66 \text{ cm}$$

$$\text{average rate} = \frac{66-50}{4} \\ = 4 \text{ cm/min}$$

26. The number of bats, N , in the park at time t years, is given by the formula
 $N = 40 \times 3^{0.327t}$.

- (a) How many bats were in the park initially?

1

$$t=0 \\ N = 40 \times 3^0 \\ = 40 \text{ bats}$$

- (b) Make t the subject of the formula.

2

$$N = 40 \times 3^{0.327t}$$

$$3^{0.327t} = \frac{N}{40}$$

$$0.327t = \log_3 \frac{N}{40}$$

$$t = \frac{\log_3 \frac{N}{40}}{0.327}$$

- (c) How many years would it take for the number of bats to reach 80? Answer to the nearest year.

2

$$t = \frac{\log_3 \frac{80}{40}}{0.327}$$

$$= \log_3 2 \times \frac{1}{0.327}$$

$$= \frac{\log 2}{\log 3} \times \frac{1}{0.327}$$

$$= 1.93$$

$$\approx 2 \text{ years}$$

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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